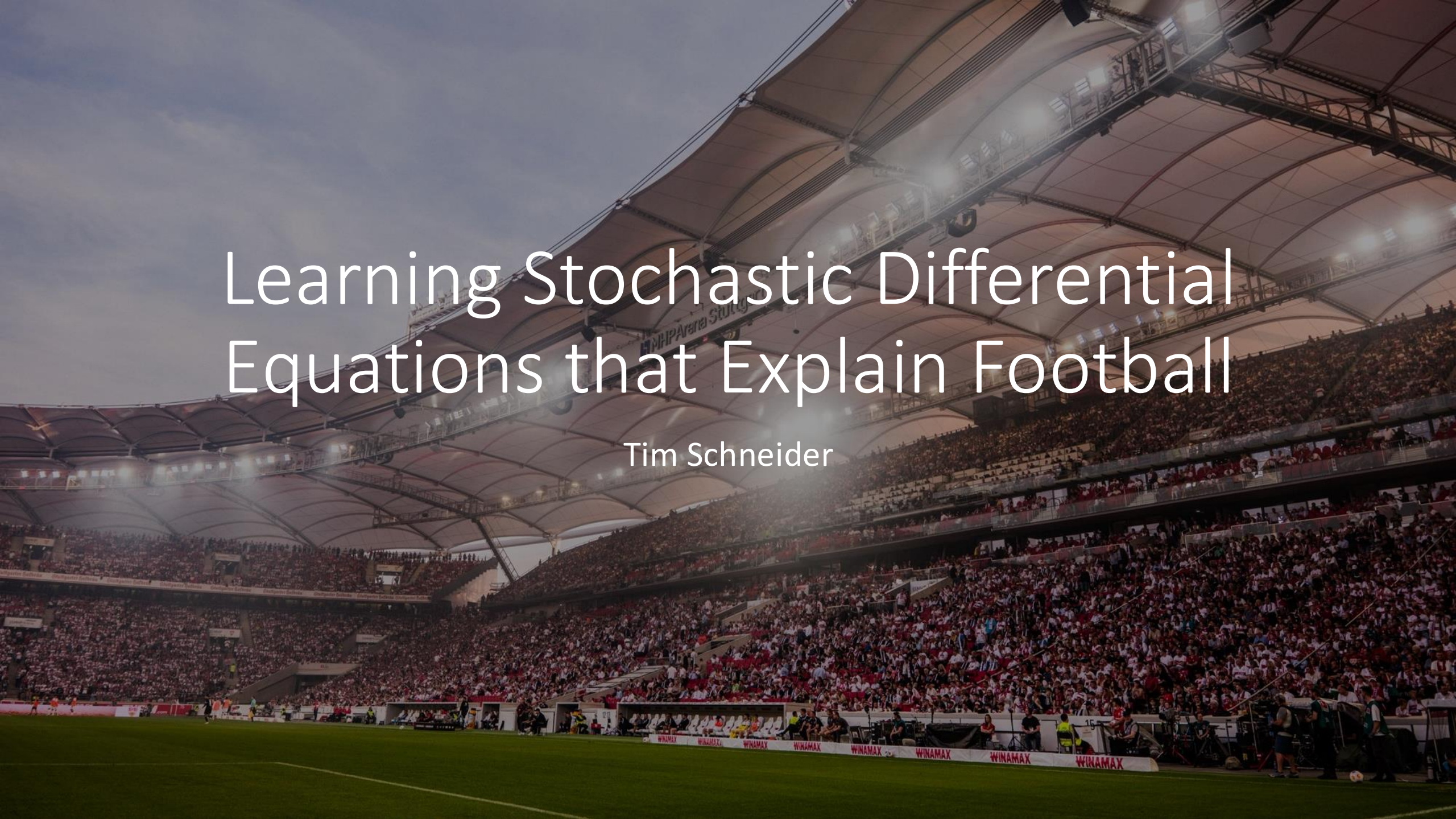


Learning Stochastic Differential Equations that Explain Football

Tim Schneider

How do football players move?

- Football is a complex and dynamic game
- The movement of players is **not deterministic**
- The movement of players is **not random**
- Can we predict players movement with **stochstic differential equations** and Machine Learning?

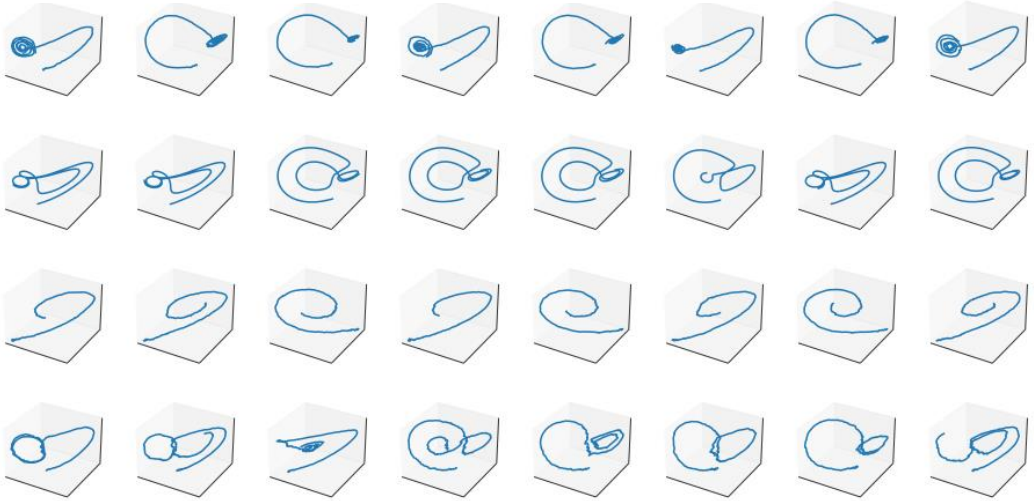
What are Stochastic Differential Equations?

(Example: Stochastic Lorenz System)

Difussion Term
(= noise part)

$$
\begin{aligned}
\mathrm{d}X_t &= \sigma(Y_t - X_t)\,\mathrm{d}t + \boxed{\alpha_x\,\mathrm{d}W_t,} & X_0 &= x_0,\\
\mathrm{d}Y_t &= (X_t(\rho - Z_t) - Y_t)\,\mathrm{d}t + \alpha_y\,\mathrm{d}W_t, & Y_0 &= y_0,\\
\mathrm{d}Z_t &= \boxed{(X_tY_t - \beta Z_t)\,\mathrm{d}t} + \alpha_z\,\mathrm{d}W_t, & Z_0 &= z_0.
\end{aligned}
$$

Drift Term
(= deterministic part)



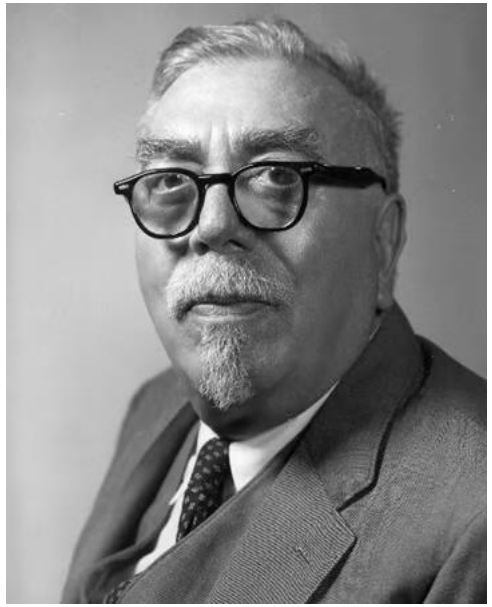
Wiener Process

- White noise independent of the time discretization

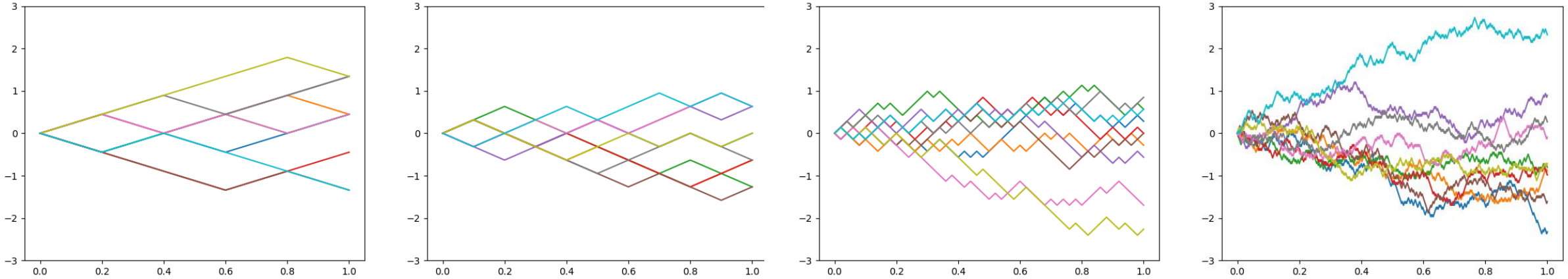
$$\text{random walk } S_n = \sum_{i=1}^{n} X_i \text{ with } p(X_i = \pm 1) = \frac{1}{2}$$

$$W_n(t) = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor}, \quad t \in [0, 1]$$

$$\text{limit } n \to \infty \text{ yields } W(t) = \mathcal{N}(0, \sqrt{t})$$

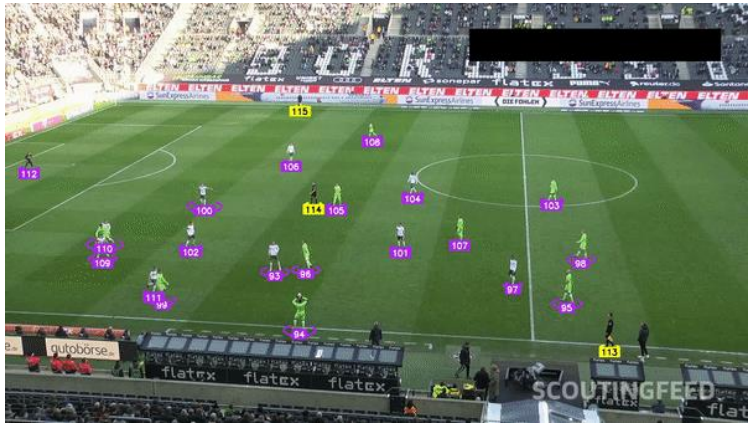


Norbert Wiener
(1894 - 1964)

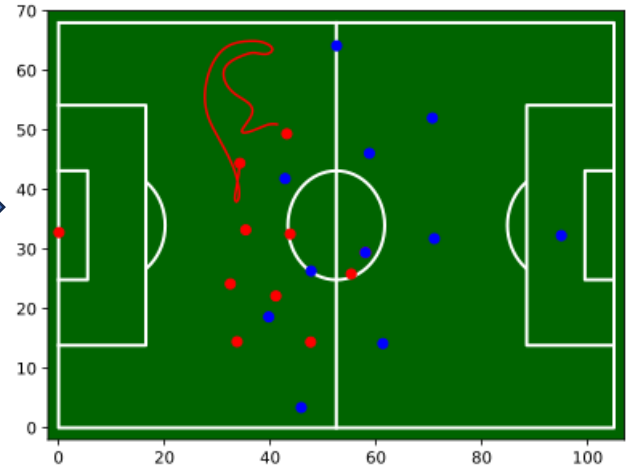


The Metrica Sports Dataset

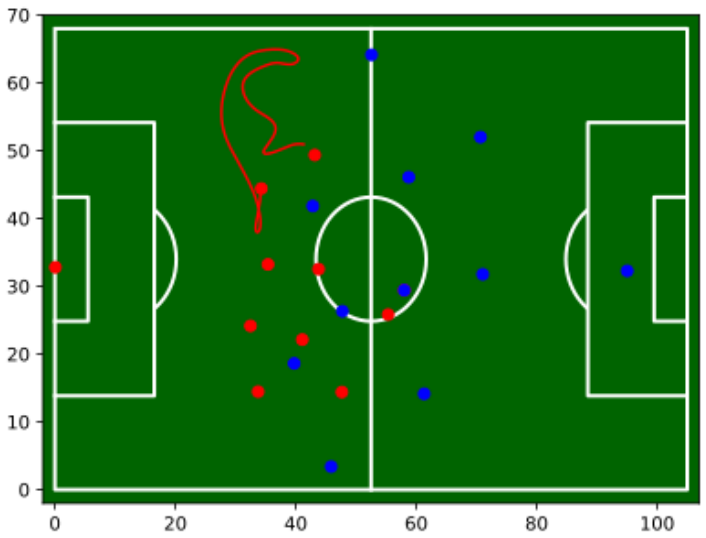
- three games anonymized
  - no references to the names of players, teams or competitions.



(also ML but not part of the task)



Examples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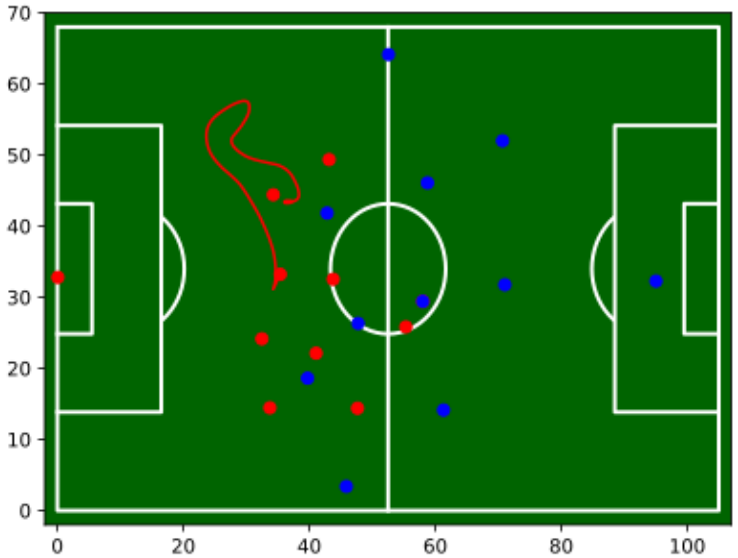
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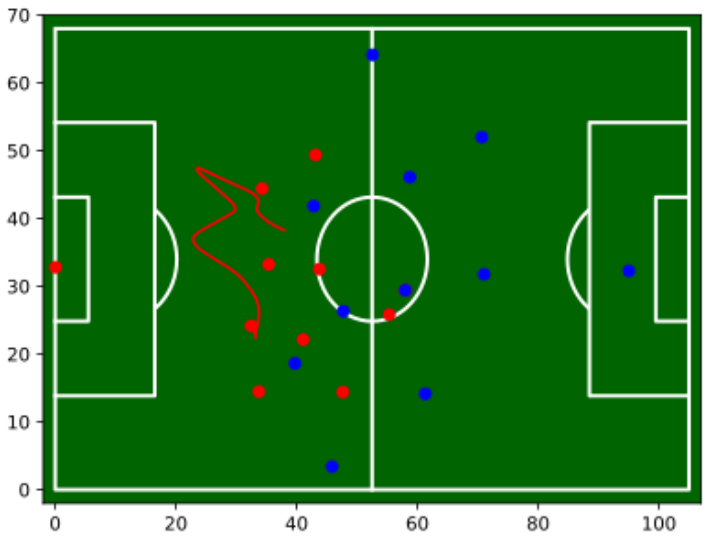
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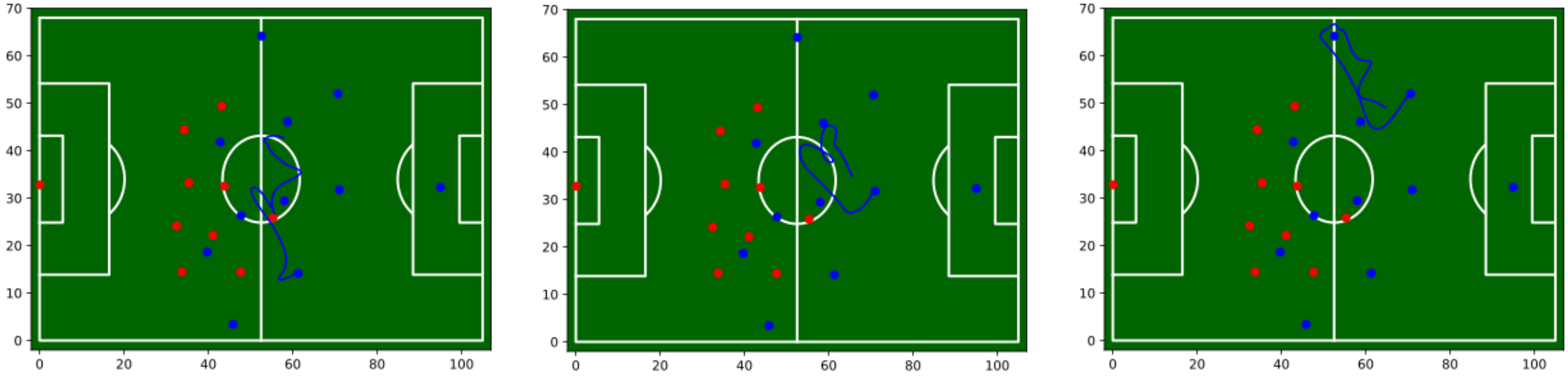
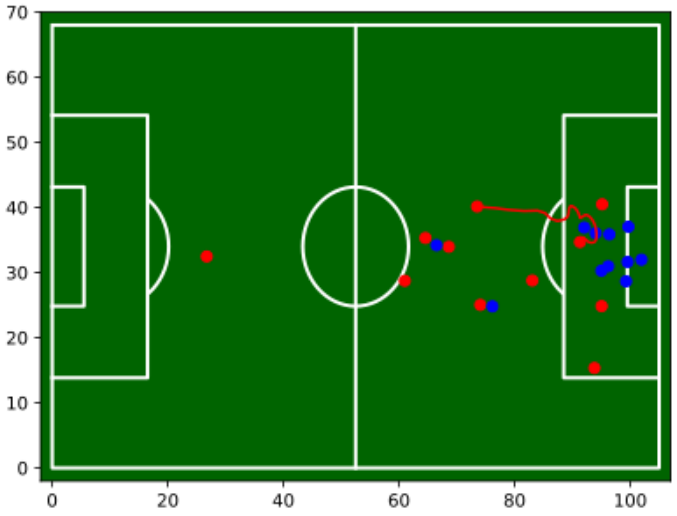
Examples



Examples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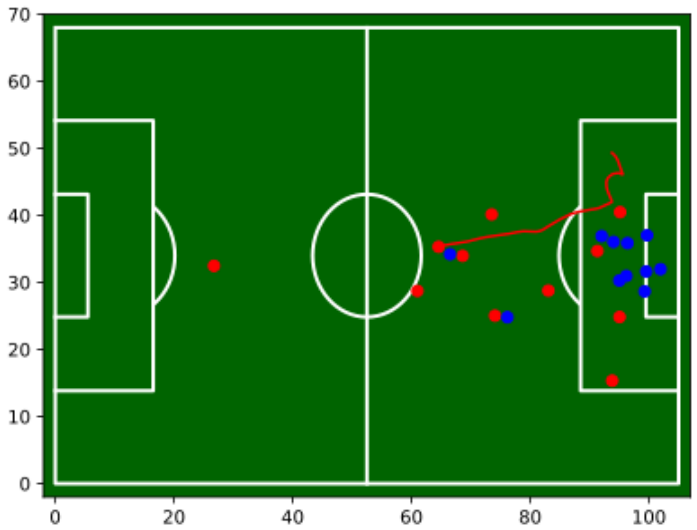
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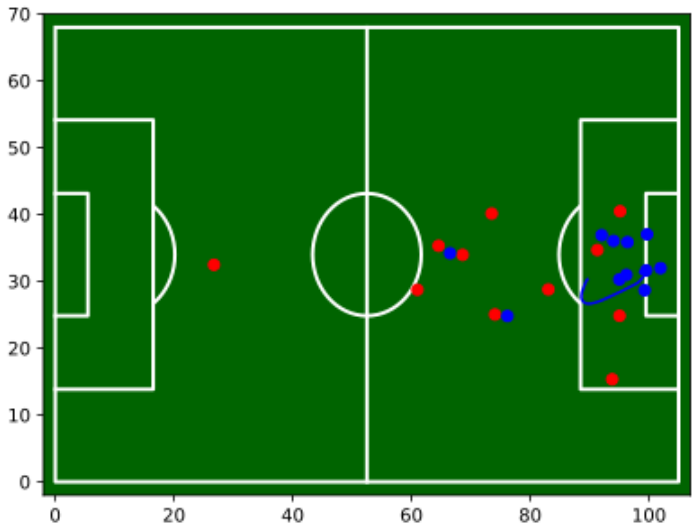
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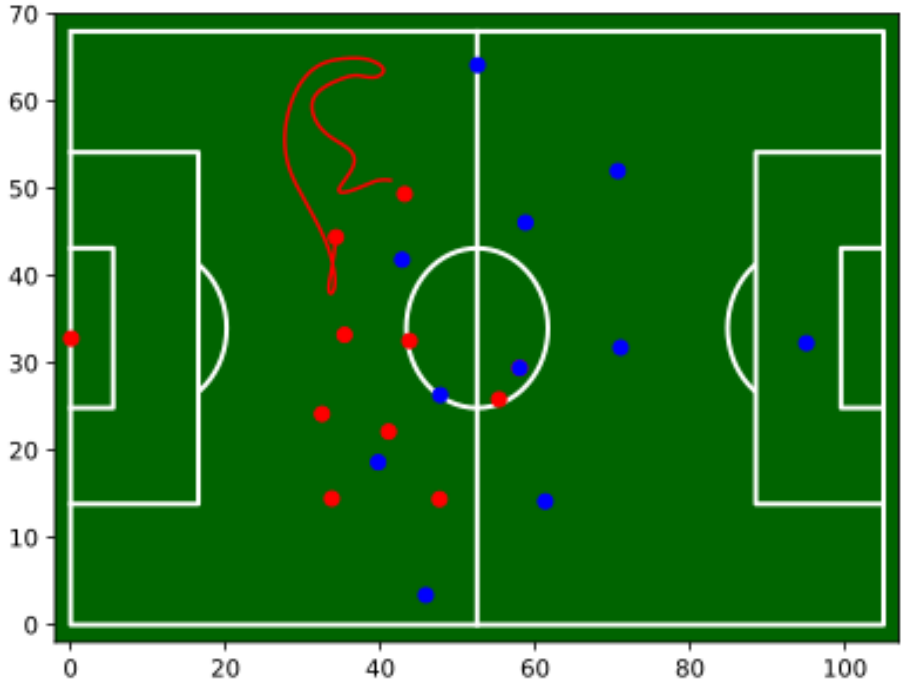
Project Overview

- Goal: Learn an equation that (somehow) explains football.

- Develop a (naive) symbolic hypothesis set
- **Task 1:** Learning of Parameters (= reproduce)
- **Task 2:** Learning to Select between Models (= novel)

Model Hypothesis Set

$$dx_0 = \alpha_0 x_0 + \beta_0 x_1 dt + dW_t$$
$$dx_1 = \alpha_1 x_0 + \beta_1 x_1 dt + dW_t$$



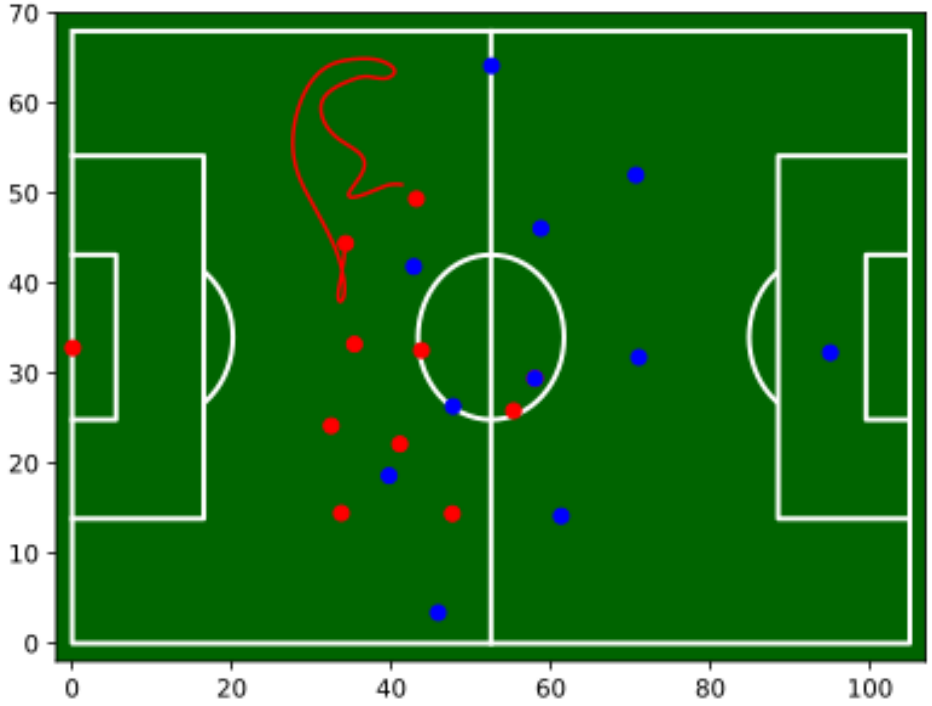
Where is the Machine Learning ?

➔ Parameters

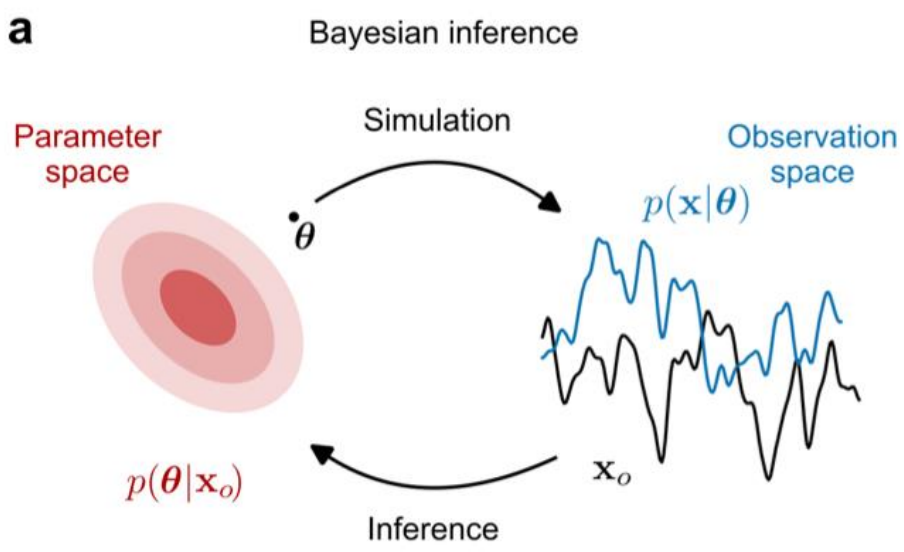
➔ What else influences the player?

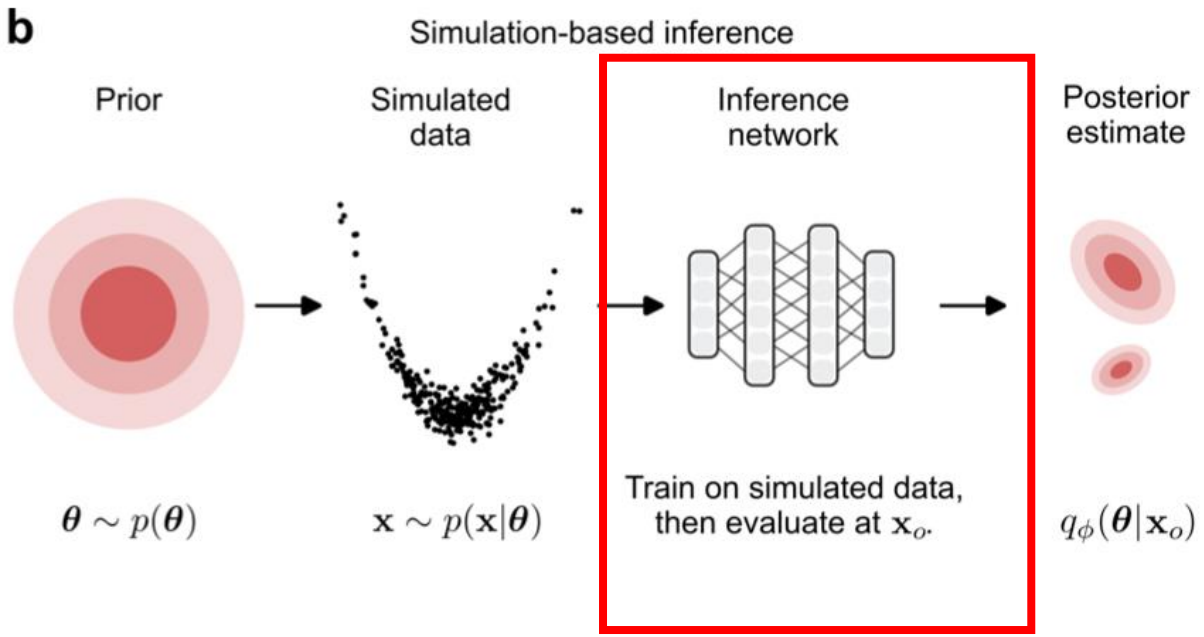
Model Hypothesis Set - Influence Factors

- (relative) Position of the ball
- (realtive) Position of teammates
- (relative) Position of opponents
- role of the player in the team
- Many other …



Simulation-Based Inference





How do we train this network?

Training with Classification

$$p(\boldsymbol{\theta} \,|\, \mathbf{x}) = \frac{p(\boldsymbol{\theta}) p(\mathbf{x} \,|\, \boldsymbol{\theta})}{p(\mathbf{x})}$$

$$p(\mathbf{x} \,|\, \boldsymbol{\theta}) = \int \delta(\mathbf{x} - \mathbf{g}(\boldsymbol{\theta}, \mathbf{z}))\, p(\mathbf{z} \,|\, \boldsymbol{\theta})\, \mathrm{d}\mathbf{z},$$

$$r(\mathbf{x}, \boldsymbol{\theta}) = \frac{p(\mathbf{x}, \boldsymbol{\theta})}{p(\mathbf{x})\, p(\boldsymbol{\theta})} = \frac{p(\mathbf{x} \,|\, \boldsymbol{\theta})}{p(\mathbf{x})} = \frac{p(\boldsymbol{\theta} \,|\, \mathbf{x})}{p(\boldsymbol{\theta})}.$$

Key insight: Optimal classifiers learn probability ratios !

$$(\mathbf{x}, \boldsymbol{\theta}) \sim p(\mathbf{x}, \boldsymbol{\theta})$$

$$(\mathbf{x}, \boldsymbol{\theta}) \sim p(\mathbf{x})\, p(\boldsymbol{\theta})$$

Neural Network

$$p(y = 1 \,|\, \mathbf{x}, \Theta) = \frac{\exp(f_{\boldsymbol{\phi}}(\boldsymbol{\theta}_1, \mathbf{x}))}{\exp(f_{\boldsymbol{\phi}}(\boldsymbol{\theta}_0, \mathbf{x})) + \exp(f_{\boldsymbol{\phi}}(\boldsymbol{\theta}_1, \mathbf{x}))},$$

References (to start with)

[Submitted on 4 Nov 2019 (v1), last revised 2 Apr 2020 (this version, v3)]

The frontier of simulation-based inference

Kyle Cranmer, Johann Brehmer, Gilles Louppe

https://arxiv.org/abs/1911.01429

On Contrastive Learning for Likelihood-free Inference

Conor Durkan [1] Iain Murray [1] George Papamakarios [2]

https://proceedings.mlr.press/v119/durkan20a/durkan20a.pdf

Likelihood-free MCMC with Amortized Approximate Ratio Estimators

Joeri Hermans, Volodimir Begy, Gilles Louppe *Proceedings of the 37th International Conference on Machine Learning*, PMLR 119:4239-4248, 2020.

https://proceedings.mlr.press/v119/hermans20a.html

Datasets

- https://www.kaggle.com/datasets/ekrembayar/metrica-sports-football-tracking-data
- https://github.com/metrica-sports/sample-data

Alternative:

https://github.com/Friends-of-Tracking-Data-FoTD/LaurieOnTracking

First Steps

- https://github.com/google-research/torchsde